

HAT-P-32b

R-0180

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_241005 · created 2026-07-16T19:09:56+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460588.8282 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.029
Transit depth (Rp/Rs)^2	1.78 +/- 0.77 [%]
Orbital Inclination (inc)	84.05 +/- 1.75
Airmass coefficient 1 (a1)	0.72 +/- 0.015
Airmass coefficient 2 (a2)	0.31 +/- 0.12
Scatter in the residuals of the lightcurve fit is	2.04 %
Best Comparison Star	#1 - [344, 291]
Optimal Aperture	4.74
Optimal Annulus	9.04
Transit Duration (day)	0.1094 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.133 $\pm$ 0.029 vs 0.1489 (deviation 0.55 σ)
Duration fit vs published	0.1094 d vs 0.1304 d (deviation 3.18 σ)
Mid-time O-C	11.3 min
Depth SNR	4.6
Residual scatter	2.04 %

Light curve

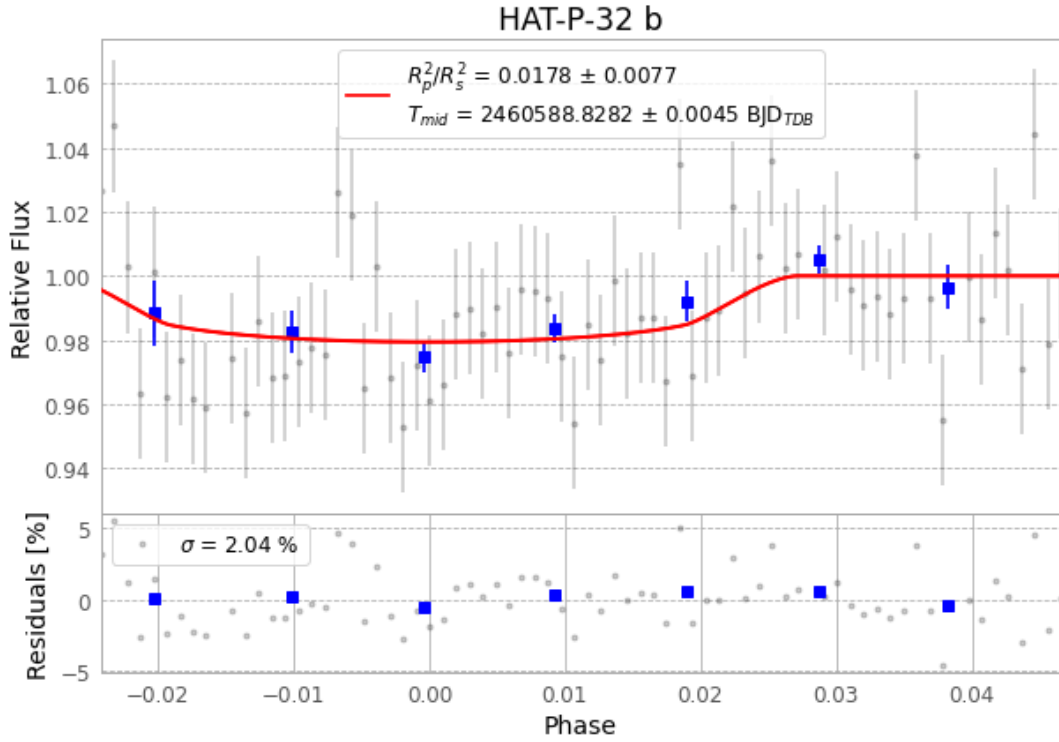


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-10-04.png (SHA-256 df75828d08ebd1a1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 235], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 22ad735ee9e51086...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca

Data provenance

75 FITS frames (49 MB), HATP-32241005062715.FITS → HATP-32241005100915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241005.log (03325840c65d...), FinalLightCurve_HAT-P-32 b_2024-10-04.png (df75828d08eb...), FinalLightCurve_HAT-P-32 b_2024-10-04.csv (74558be7796c...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:09Z.